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Contrastive Representation Learning on High-Dimensional Scientific Embedding Manifolds

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ABSTRACT

Embedding scientific literature requires capturing multi-faceted semantic relationships ranging from domain taxonomy to methodology overlap. We develop a multi-scale contrastive learning framework that trains embeddings across joint document, citation-neighborhood, and entity-level manifolds. By employing hard-negative mining based on false-citation pairs, our model prevents representation collapse in specialized scientific vocabularies. Evaluations on scientific document clustering and nearest-neighbor search yield a 16.4% gain in Mean Reciprocal Rank over standard SciBERT embeddings. Nonetheless, hyperparameter sensitivity in hard-negative temperature scaling requires careful cross-validation to avoid cluster fragmentation.

Keywords— Machine Learning, Contrastive Learning, Scientific Embeddings, Representation Learning

INTRODUCTION

Scientific text embeddings often suffer from topic blurring because papers in identical sub-fields share dense vocabulary despite proposing diametrically opposed methodologies. We propose a multi-scale contrastive objective to disentangle these nuances.

METHODOLOGY

We define hierarchical contrastive losses: document-level SimCLR loss, neighborhood preservation loss on the citation graph, and token-level masked entity alignment. Hard negatives are actively mined from uncited papers with high lexical overlap.

RESULTS

On the SciDocs benchmark, our representations achieved highest clustering purity (82.4%) and top retrieval precision, consistently separating foundational methods from minor incremental extensions.

LIMITATIONS

The hard-negative temperature parameter exhibits high sensitivity; a 0.02 deviation in temperature can cause embedding space collapse or excessive metric fragmentation on rare topics.

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